

Economic Denial of Sustainability (EDoS) attack detection by attention on flow-based in Software Defined Network (SDN)

1st Quoc Vinh Ta

Department of Information Communication, Materials and Chemistry Convergence Technology Soongsil University.
taquocvinh@soongsil.ac.kr

2nd Minh Park

Department of Information Communication, Materials and Chemistry Convergence Technology Soongsil University.
School of Electronic Engineering, Soongsil University, Seoul 156-743, South Korea.
mhp@ssu.ac.kr

Abstract—Economic Denial of Sustainability (EDoS) attack, a kind of Denial of Service attack (DoS), is one of the potential threats to cloud computing. Since the characteristics of EDoS attacks such as stealthiness and long-term period make EDoS detection and prevention challenging, various detection mechanisms have been proposed. However, they mainly focused on utilizing the features of EDoS attack traffic to enhance detection performance and overlooked the environment where the defense mechanisms are running over. We realized exiting proposals have limitations, such as unnecessary overhead or performance degradation since they do not exploit the features provided in an SDN-based cloud. To overcome the limitations, we propose an efficient EDoS detection system optimized for SDN-based clouds. It applies the attention technique to calculate the attention scores of the flow network extracted from the SDN controller to detect the EDoS attack. The performance evaluation results show that it achieves better accuracy and processing time than previous work.

Index Terms—Cloud computing, economic denial of sustainability (EDoS), software-defined network, machine learning, attention, flow-based characteristic.

I. INTRODUCTION

Cloud computing offers many advantages in various ways. One of the advantages is that it allows users to exploit the computing resource based on the pay-per-use service irrespective of time and place. As cloud computing is getting pervasive due to its advantages, the risk of diverse threats is also increasing. The Economic Denial of Sustainability (EDoS) attack is one of the cloud-specific attacks. It consumes the cloud resource maliciously and leads a victim to bankruptcy due to an excessive bill.

Although EDoS is one type of Denial of Service (DoS) attack, they have different purposes. While the DoS attack aims to prevent a victim from providing service immediately, the EDoS attack does not stop the victim's service. It mimics normal clients and gradually exhausts the computing resources for a long time, which is the unique characteristic of EDoS attacks. The threshold of attack rate is lower than the original DoS attack and slightly higher than the normal traffic. This

stealthiness makes a defense system difficult and challenging to detect EDoS.

By far, there have been proposed various mechanisms based on statistics or machine learning to detect and prevent EDoS. However, these methods are limited in the detection performance because they focus on utilizing the features of EDoS attack traffic to gain accuracy in detection performance but do not consider the network infrastructure. Although Software-Defined Network (SDN), the most common network infrastructure for cloud computing, provides the useful information for network flows, the existing methods do not use the information but mainly analyze traffic in a packet-based manner.

Due to the separation of the data plane and control plane, the SDN significantly benefits in optimizing the EDoS defense system. The advantage of SDN architecture in contrast with traditional networking architecture is that SDN allows developers to manage and configure all the network devices remotely. In SDN, the data plane includes the forwarding devices such as routers and switches, and they forward packets based on flow tables populated by the control plane. Besides, SDN provides the developers to calculate the unique flow features through the OpenFlow protocol between switches and the controller. These flow features are considered to be crucial for the EDoS detection issues.

Thus, we proposed an efficient EDoS detection system optimized for SDN-based clouds computing by applying the flow-based features. Moreover, we proposed the attention technique inspired by the Transformer model to compute the relevant scores of the flow features extracted from the flow table of the SDN controller. These scores improve the system to track the changing of the flow features in order to detect the EDoS detection flow. The performance evaluation shows the proposed system gained better results in multi-metric than the previous works. Our work reduces the false alarm rate and gains better accuracy than statistical methods. In addition, the parallel computation of attention layers illustrated that the processing time of attention model is faster than the processing

time of recurrent models because recurrent models wasted so much on gradual training steps.

The rest of the paper compiles as follow: we introduce some related works, and its limitation in section II. The proposed scheme is described in section III. The evaluations and results are presented in section IV. Section V concludes the paper and presents the future research directions.

II. RELATED WORKS

This proposed scheme is inspired by a combination of conclusions from existing researches that worked on the EDoS detection listed below:

- **EDoS Armor** [2]: The EDoS Armor is a packet filtering and authentication method based on the composition of the admission control phase and the congestion control phase. However, this work is not easy to adapt, and it provides a defense system only for E-commerce applications.
- **EDoS Shield** [3]: The shield is established on the virtual firewalls and the verified cloud nodes to contain the IP black and white lists in order to accept or reject the packets incoming. The limitation of the EDoS Shield is that it does not deal with IP spoofing and become vulnerable to exploitation.
- **sPoW** [4]: self-verifying proof of work is an on-demand cloud-based and application layer mitigation technique. This technique applied packets filtering and crypto-puzzle to deal with the EDoS packets detection. On the other hand, this work is prevented only on the network-level.
- **Machine Learning** [5]: The usual method for Machine Learning algorithms for EDoS is Support Vector Machine (SVM). Nevertheless, the SVM is based on the statistical of the network packets to calculate the most distance vector to the convergence boundary of classes. With the development of the network data, the more massive amount of data increases, the worse performance of the SVM shows off. These varieties of data come with the vulnerability in threshold detection for EDoS attacks.
- **Long-Short Term Memory** [6], [7]: The recurrent networks that perform in the sequence data have the ability to track the changes in values of the flow data instead of the packets classification as previous methods as illustrated in previous works, recurrent or LSTM especially model that had higher accuracy and higher scores of metrics than previous novels. However, as good performance in the EDoS detection as recurrent networks, they still got some limitations in the memory stacking and time processing.

To overcome the limitation of the recurrent model in flow-based applied for EDoS detection in SDN, our proposed introduced the attention techniques to compute the attention scores between flow feature extracted from the flow table in SDN controller. This research is the development from [8] despite the enhancement from packet-based calculation to flow-based extraction from flow table of SDN. Throughout our

experiment, attention technique that is extraordinary in EDoS classification issue with flow-based features extracted.

III. PROPOSED MECHANISM AND EXPERIMENT

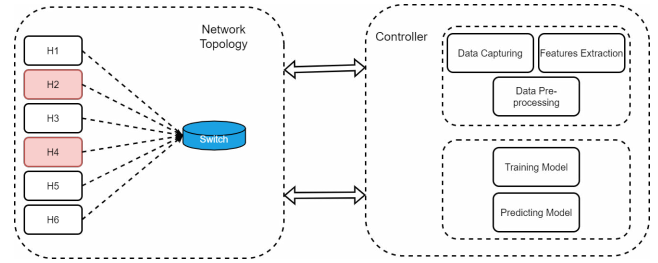


Fig. 1. SDN topology for EDoS detection system.

In the experiment, we constructed the SDN with controller and network topology to simulate the EDoS attack and flow extraction from the flow table in the controller as in Figure 1. As the topology figure, the hosts or the clients are connected to each other throughout the switch and through the open flow interface connected to the controller. The SDN controller has much administrations to control the flows coming and access the flows to the flow table for standard rules in order to bypass them or prevent them from connecting to the other hosts. The flows are then pre-processed to normalize the inappropriate values and categories data to feed to the model. The next

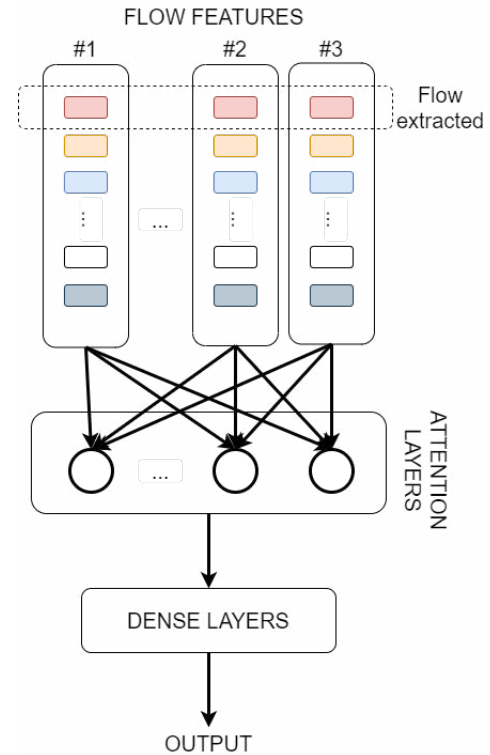


Fig. 2. Model architecture for attention on EDoS attacks.

step after data processing is feeding into the model the train the distinguished threshold. The figure 2 illustrated the model

architecture of the attention network. The attention layers are calculated as in [9] and then being fed to the basic neuron net for making the decision score output. The attention scores are calculated by constructing 3 latent matrices query, key, and value matrices. The query and key matrices are represented for the flow features relation in the network flow and then are scored by the value matrices to generate the attention matrices. By these matrices, the proposed model is able to follow the changes in values of the features in the flow.

IV. RESULTS AND EVALUATION

By training of 100 epochs with ADAM optimizer, binary cross-entropy loss function to converge.

The figure 3 and 4 above show that our proposal work

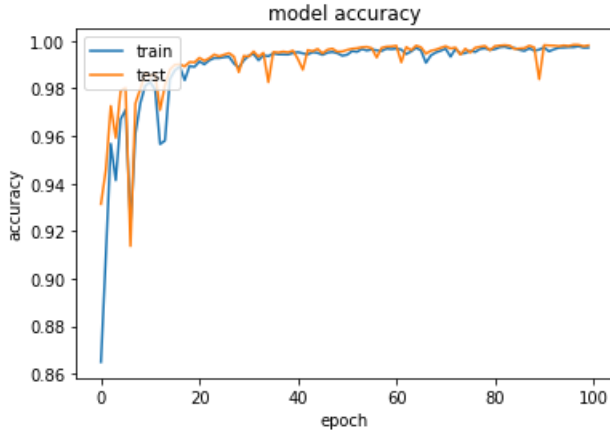


Fig. 3. The accuracy of attention model on EDoS detection.

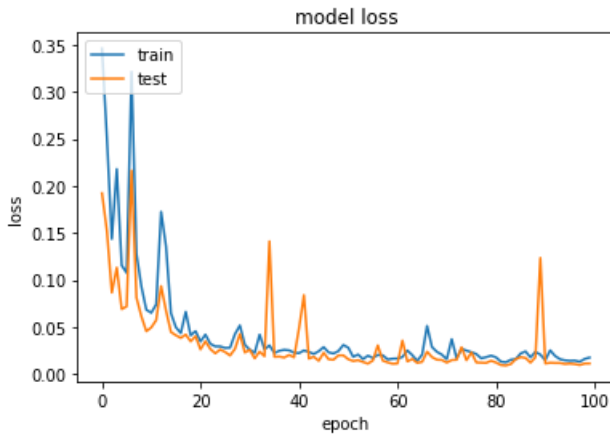


Fig. 4. The loss of attention model on EDoS detection.

gained very significant results in accuracy scores of the training process. By comparison, our proposal gained to the peak of 99% of accuracy in EDoS detection system in SDN environment. As a result, our results are better than the recent works of EDoS detection that applied recurrent models or any statistical mechanisms.

The table I above shows the results of the comparison of

TABLE I
RESULTS COMPARISON

Metrics	Our Proposal	Recurrent models
Accuracy	0.9898	0.979
Time processing (s/epochs)	11	50

our proposal with the previous recurrent models, in this case is RNN and LSTM models. As the illustration, our results are better at accuracy and faster in processing time than previous novels.

V. CONCLUSION

To sum up, our proposal is about the EDoS detection and prevention system by applying the flow-based method. Moreover, by applying the attention technique to compute the focused scores in the flow features in the SDN environment, our proposal has gained better results than other novels. By the attention scores, we can improve the scheme to a better system by adjusting the optimization and process methods, these modules are helped by the SDN controller characteristics.

ACKNOWLEDGEMENT

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT) (No. 2020R1F1A1076795).

REFERENCES

- [1] Wadhwa S., Mandhar V. (2021) Survey on DDoS and EDoS Attack in Cloud Environment. In: Singh P.K., Noor A., Kolekar M.H., Tanwar S., Bhatnagar R.K., Khanna S. (eds) *Evolving Technologies for Computing, Communication and Smart World. Lecture Notes in Electrical Engineering*, vol 694. Springer, Singapore. https://doi.org/10.1007/978-981-15-7804-5_24
- [2] Masood, M.; Anwar, Z.; Raza, S.A.; Hur, M.A. EDoS Armor: A cost effective economic denial of sustainability attack mitigation framework for e-commerce applications in cloud environments. In *Proceedings of the INMIC, Lahore, Pakistan, 19–20 December 2013*; pp. 37–42.
- [3] Al-Haidari, F.; Salah, K.; Sqalli, M.; Buhari, S.M. Performance Modeling and Analysis of the EDoS-Shield Mitigation. *Arab. J. Sci. Eng.* 2017, 42, 793–804.
- [4] Khor, S.H.; Nakao, A. Spow On-Demand Cloud-based EDDoS Mitigation Mechanism. In *Proceedings of the 5th Workshop on Hot Topics in System Dependability, Lisbon, Portugal, 29 June 2009*; pp. 1–6.
- [5] Shideh Yavary Mehr and Byrav Ramamurthy. 2019. An SVM Based DDoS Attack Detection Method for Ryu SDN Controller. In *Proceedings of the 15th International Conference on emerging Networking EXperiments and Technologies (CoNEXT '19 Companion)*. Association for Computing Machinery, New York, NY, USA, 72–73. DOI:<https://doi.org/10.1145/3360468.3368183>
- [6] Dinh, P.T.; Park, M. Dynamic Economic-Denial-of-Sustainability (EDoS) Detection in SDN-based Cloud. In *Proceedings of the 2020 Fifth International Conference on Fog and Mobile Edge Computing (FMEC)*, Paris, France, 20–23 April 2020; pp. 62–69.
- [7] Li, Y.; Lu, Y. LSTM-BA: DDoS Detection Approach Combining LSTM and Bayes. In *Proceedings of the 2019 Seventh International Conference on Advanced Cloud and Big Data (CBD)*, Suzhou, China, 21–22 September 2019; pp. 180–185.
- [8] Ta, V.Q.; Park, M. MAN-EDoS: A Multihead Attention Network for the Detection of Economic Denial of Sustainability Attacks. *Electronics* 2021, 10, 2500. <https://doi.org/10.3390/electronics10202500>
- [9] Ashish Vaswani; Noam Shazeer; Niki Parmar; Jakob Uszkoreit; Llion Jones; Aidan N. Gomez; Lukasz Kaiser; Illia Polosukhin. Attention Is All You Need. *arXiv* 2017, arXiv: 1706.03762.